MTH-418: Inference-I

2025-2026: II Semester
Mid Semester Examination

Time Allowed: 120 Minutes **Maximum Marks: 50**

NOTE: (i) Start answer of every question on a new page. Moreover, attempt all the parts of a question at one place.
(ii) Answer each question legibly, clearly and concisely. Illegible answers will not be graded. For each question, provide details of all the steps involved in arriving at the final answer(s)/conclusion(s) and box your final answer(s)/conclusion(s).

1. Let X_1, X_2, X_3 be a random sample of size 3 from a discrete uniform distribution on the set $\{1, 2, \ldots, \theta\}$, where $\Theta = \{2, 3, \ldots\}$ (the parameter space Θ is restricted in the sense that $1 \notin \Theta$). Let $X_{(1)} = \min\{X_1, X_2, X_3\}$ and $X_{(3)} = \max\{X_1, X_2, X_3\}$.

   (a) Show that $X_{(3)}$ is sufficient but **NOT** complete.

   (b) Find the UMVUE of $\psi(\theta) = \frac{1}{\theta}$. $\boxed{\text{6+6=12 Marks}}$

2. Let X_1, X_2, X_3 be a random sample from $\mathrm{U}(\theta_1 - \theta_2, \theta_1 + \theta_2)$ distribution, where $\underline{\theta} = (\theta_1, \theta_2) \in \{(x, y) \in \mathbb{R}^2 : -\infty < x < \infty, y > 0\} = \Theta$, say.

   (a) Find the MME of (θ_1, θ_2).

   (b) Find the MLE of (θ_1, θ_2). $\boxed{\text{6+7=13 Marks}}$

3. Let X_1, X_2, X_3 be a random sample from $\mathrm{Exp}(\theta)$ distribution, where $E_\theta(X_1) = \theta \in (0, \infty) = \Theta$, say. For estimating θ, under the squared error loss function $L(\theta, a) = (a - \theta)^2$, $\theta \in \Theta, a \in (0, \infty) = \mathcal{A}$, say, consider the randomized estimator $\delta^{(1)}_{\underline{X}} \sim \mathrm{U}(0, X_1 + X_2)$. Let $T = X_1 + X_2 + X_3$.

   (a) Show that T is a complete-sufficient statistic.

   (b) Find an estimator based on T having the same risk function as $\delta^{(1)}_{\underline{X}}$.

   (c) Find a non-randomized estimator based on T that has smaller risk values than those of $\delta^{(1)}_{\underline{X}}$. $\boxed{\text{3+4+3+3=13 Marks}}$

   (d) For real constants a and b, find the UMVUE of $\psi(\theta) = a\theta + b\theta^2$.

4. Let X_1, X_2, X_3 be a random sample from $\mathrm{U}(0, \theta)$ distribution, where $\theta \in (0, \infty) = \Theta$, say. Assume that $X_{(3)} = \max\{X_1, X_2, X_3\}$ be a complete-sufficient statistic.

   (a) Using Basu's theorem, find the conditional distribution of X_1, given $X_{(3)} = 2$.

   (b) Find the value of $E_\theta(\frac{X_{(1)}}{X_{(3)}})$, where $X_{(1)} = \min\{X_1, X_2, X_3\}$. $\boxed{\text{6+6=12 Marks}}$

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Model Solutions

Problem No. 1 (a) The joint pdf of $\underline{X} = (X_1, X_2, X_3)$ is

$$g_\theta(\underline{x}_1, \underline{x}_2, \underline{x}_3) = G(x_{(3)}, \theta)\, h(\underline{x}), \quad \underline{x} \in \mathbb{R}^3$$

where $G(t, \theta) = \frac{1}{\theta^3} I(t \in \{1, \dots, \theta\})$ and $h(\underline{x}) = I(x_{(1)} \in \{1, 2, \dots\})$

By factorization theorem $X_{(3)}$ is sufficient. ... 3 MARKS

$$P(X_{(3)} = t) = P(X_{(3)} \le t) - P(X_{(3)} \le t-1)$$

$$= \frac{t^3 - (t-1)^3}{\theta^3}, \quad t \in \{1, 2, \dots, \theta\}, \quad \theta \in \Theta = \{2, 3, \dots\}$$

Let $h(\cdot)$ be such that

$$E_\theta(h(X_{(3)})) = 0 \quad \forall\, \theta \in \Theta = \{2, 3, \dots\}$$

$$\Leftrightarrow \sum_{t=1}^{\theta} h(t)\,[t^3 - (t-1)^3] = 0, \quad \forall\, \theta = 2, 3, \dots$$

$\theta = 2$: $\quad h(1) + 7h(2) = 0$

$$\sum_{t=1}^{\theta-1} h(t)\,[t^3 - (t-1)^3] = 0 \quad \forall\, \theta = 3, 4, \dots$$

$$\Rightarrow h(\theta)\,[\theta^3 - (\theta-1)^3] = 0 \quad \forall\, \theta = 3, 4, \dots$$

$$\Rightarrow h(2) = -\frac{h(1)}{7}, \quad h(t) = 0, \quad t = 3, 4, \dots$$

One may take

$$h(1) = 1, \quad h(2) = -\frac{1}{7}, \quad h(t) = 0 \quad t = 3, 4, \dots$$

Then $E_\theta[h(X_{(3)})] = 0 \quad \forall\, \theta \in \Theta = \{2, 3, \dots\}$

but $P_\theta(h(X_{(3)}) = 0) = P_\theta(X_{(3)} \in \{3, 4, \dots, \theta\}) < 1 \quad \forall\, \theta \in \Theta$

$\Rightarrow X_{(3)}$ is NOT complete ... 3 MARKS

(b) For finding the UMVUE, it suffices to consider estimators that depend on $\underline{X}$ only through $X_{(3)}$ (minimal sufficient statistic).

$$\delta \in \Theta_U \iff E_\theta(\delta(X_{(3)})) = \frac{1}{\theta}, \quad \forall\, \theta \in \{2, 3, \dots\}$$

$$\iff \sum_{t=1}^{\theta} \delta(t)\,[t^3 - (t-1)^3] = \theta^2, \quad \forall\, \theta \in \{2, 3, \dots\}$$

$\theta = 2$: $\quad \delta(1) + 7\delta(2) = 4$

$$\sum_{t=1}^{\theta-1} \delta(t)\,[t^3 - (t-1)^3] = (\theta-1)^2, \quad \theta \in \{3, 4, \dots\}$$

$$\Rightarrow \quad \delta(\theta)\,[\theta^3 - (\theta-1)^3] = \theta^2 - (\theta-1)^2, \quad \theta = 3, 4, \dots$$

$$\iff \boxed{\begin{aligned} &\delta(1) + 7\delta(2) = 4 \\ &\delta(\theta) = \frac{2\theta - 1}{3\theta^2 - 3\theta + 1}, \quad \theta = 3, 4, \dots \end{aligned}} \quad \dots (I)$$

... 3 MARKS

Any δ satisfying above will be the UMVUE of $\psi(\theta) = \frac{1}{\theta}$

iff $\quad E_\theta(\delta(X_{(3)})\, U(X_{(3)})) = 0, \quad \forall\, \theta \in \{2, 3, \dots\}, \quad \forall\, U \in \mathcal{U}$

$$\iff \delta(1)\,U(1) + 7\delta(2)\,U(2) = 0$$

and $\quad \delta(t)\,U(t) = 0, \quad \forall\, t \in \{3, 4, \dots\}$

and $\quad \forall\, U(\cdot)$ satisfying

$$U(1) + 7U(2) = 0, \quad U(t) = 0, \quad \forall\, t \in \{3, 4, \dots\}$$

$$\iff (\delta(1) - \delta(2))\,U(1) = 0,$$

$$\iff \delta(1) = \delta(2).$$

Using in (I) the UMVUE of $\psi(\theta) = \frac{1}{\theta}$ is obtained as

$$\delta(X_{(3)}) = \begin{cases} \frac{1}{2}, & X_{(3)} = 1, 2 \\ \dfrac{2X_{(3)} - 1}{3X_{(3)}^2 - 3X_{(3)} + 1}, & X_{(3)} = 3, 4, \dots \end{cases}$$

... 3 MARKS

Problem No. 2

(a) $E_{\underline{\theta}}(X_1) = \theta_1$,

$E_{\underline{\theta}}(X_1^2) = \dfrac{3\theta_1^2+\theta_2^2}{3}$

$\Rightarrow \hat{\theta}_{1,MME} = \bar{X}$ and $\dfrac{3\hat{\theta}_{1,MME}^2 + \hat{\theta}_{2,MME}^2}{3} = \dfrac{1}{3}\sum_{i=1}^{3} X_i^3$,

... 4 MARKS

where $\bar{X} = \frac{1}{3}\sum_{i=1}^{3} X_i$

$\Rightarrow (\hat{\theta}_{1,MME}, \hat{\theta}_{2,MME}) = \left(\bar{X}, \sum_{i=1}^{3}(X_i-\bar{X})^2\right)$ ... 2 MARKS

(b) The likelihood function is

$$l_{\underline{x}}(\theta_1, \theta_2) = \begin{cases} \dfrac{1}{8\theta_2^3}, & \theta_2 \geq \max\{\theta_1 - x_{(1)}, x_{(3)} - \theta_1\} \\ 0, & \text{o.w.} \end{cases}$$

$$\leq \frac{1}{8\left[\max\{\theta_1 - x_{(1)}, x_{(3)}-\theta_1\}\right]^3},$$ with equality at $\theta_2 = \max\{\theta_1 - x_{(1)}, x_{(3)}-\theta_1\}$

... 3 MARKS

$$\max\{\theta_1 - x_{(1)}, x_{(3)}-\theta_1\} = \begin{cases} x_{(3)}-\theta_1, & \theta_1 < \dfrac{x_{(1)}+x_{(3)}}{2} \\ \theta_1 - x_{(1)} & \theta_1 \geq \dfrac{x_{(1)}+x_{(3)}}{2} \end{cases} \geq 0$$

$\longrightarrow$ minimized at $\theta_1 = \dfrac{x_{(1)}+x_{(3)}}{2}$

Thus

$$l_{\underline{x}}(\theta_1, \theta_2) \leq \frac{1}{(x_{(3)}-x_{(1)})^3}, \quad \forall\, \underline{\theta} \in \Theta$$

with equality at $(\theta_1, \theta_2) = \left(\dfrac{x_{(1)}+x_{(3)}}{2}, \dfrac{x_{(3)}-x_{(1)}}{2}\right)$

Thus

$$(\hat{\theta}_{1,ME}, \hat{\theta}_{2,MLE}) = \left(\frac{X_{(1)}+X_{(3)}}{2}, \frac{X_{(3)}-X_{(1)}}{2}\right)$$ ... 4 MARKS

Problem No. 3

(a) The joint pdf of $\underline{X} = (X_1, X_2, X_3)$ is

$$g_\theta(x_1, x_2, x_3) = e^{\eta(\theta)\sum_{i=1}^{3} x_i + B(\theta)} h(\underline{x}), \quad \underline{x} \in \mathbb{R}^3,$$

where $\eta(\theta) = -\frac{1}{\theta} \in (-\infty, 0)$ contains a one-dimensional rectangle

$B(\theta) = -3\ln\theta$ and $h(\underline{x}) = I(\underline{x} \in (0,\infty)^3)$.

By the result for exponential family $T = X_1 + X_2 + X_3$ is a complete-sufficient statistic. .. 3 MARKS

(b) $$\delta_{\underline{X}}^{(1)} \sim \beta_{\underline{X}}^{(1)}(a) = \frac{1}{X_1+X_2} I(X_1+X_2 > a), \quad a > 0$$

The derived estimator is

$$\delta_{\underline{X}}^{(2)} \sim \beta_{\underline{X}}^{(2)}(a) = E\left[\frac{1}{X_1+X_2} I(X_1+X_2 > a) \,\middle|\, T\right]$$

-- 2 MARKS

For fixed $t > 0$ and $a > 0$

$$E\left[\frac{1}{X_1+X_2} I(X_1+X_2 > a) \,\middle|\, T = t\right]$$

$$= \frac{1}{t} E\left[\frac{T}{X_1+X_2} I\left(\frac{X_1+X_2}{T} > \frac{a}{t}\right) \,\middle|\, T = t\right]$$

$$= \frac{1}{t} E\left[\frac{T}{X_1+X_2} I\left(\frac{X_1+X_2}{T} > \frac{a}{t}\right)\right],$$

as $\frac{X_1+X_2}{T}$ is ancillary and T is complete-sufficient (Basu's Thm). Note that wlog we may take $\theta = 1$ (as $\frac{X_1+X_2}{T}$ is ancillary) and then $\frac{X_1+X_2}{T} \sim \text{Beta}(2, 1)$. Thus

$$E\left[\frac{1}{X_1+X_2} I(X_1+X_2 > a) \,\middle|\, T = t\right] = \begin{cases} \frac{1}{t}\int_{a/t}^{1} \frac{1}{z} \cdot 2z \, dz & \frac{a}{t} < 1 \\ 0 & \frac{a}{t} > 1 \end{cases}$$

$$= \begin{cases} \frac{2}{t}\left(1 - \frac{a}{t}\right), & 0 < a < t \\ 0, & a > t \end{cases}$$

... 2 MARKS

Thus the derived estimator is

$$\delta_{\underline{X}}^{(2)} \sim p_{\underline{X}}^{(2)}(a) = \begin{cases} \frac{2}{T}\left(1-\frac{a}{T}\right), & 0<a<T \\ 0, & a>T. \end{cases}$$

(c) Since $\delta_{\underline{X}}^{(1)}$ and $\delta_{X}^{(2)}$ have the same risk functions, it suffices to find an estimator based on T that dominates $\delta_{\underline{X}}^{(2)}$. The derived estimator is the mean of $p_{\underline{X}}^{(2)}$, i.e. the derived estimator is

$$\delta^{(3)}(\underline{X}) = \int_0^T a \cdot \frac{2}{T}\left(1-\frac{a}{T}\right) da$$

$$= 2T \int_0^1 z(1-z)\, dz = \frac{T}{3} = \frac{X_1+X_2+X_3}{3}$$

... $\boxed{\text{3 MARKS}}$

(d) T is complete sufficient with

$E_\theta\left(\frac{T}{3}\right) = \theta$ and $E_\theta\left(\frac{T^2}{9}\right) = \frac{4}{3}\theta^2$ .. $\boxed{\text{2 MARKS}}$

$$E_\theta\left(\frac{aT}{3} + \frac{b}{12}T^2\right) = a\theta + b\theta^2, \quad \forall\, \theta > 0$$

By Lehmann Scheffe Theorem, $\delta(T) = \frac{aT}{3} + \frac{b}{12}T^2$ is the UMVUE of $\psi(\theta) = a\theta + b\theta^2$. $\boxed{\text{1 MARK}}$

Problem No. 4

(a) For $x < 0$, $P_\theta(X_1 \le x \mid X_{(3)} = 2) = 0$ and, for $x \ge 2$,

$P_\theta(X_1 \le x \mid X_{(3)} = 2) = 1$, $\forall\ \theta > 0$.

Also $X_{(3)}$ is complete-sufficient and $\frac{X_1}{X_{(3)}}$ is ancillary

so by Basu's Theorem, they are statistically independent.

For $0 \le x < 2$,

$$P_\theta(X_1 \le x \mid X_{(3)} = 2) = P_\theta\left(\frac{X_1}{X_{(3)}} \le \frac{x}{2} \,\middle|\, X_{(3)} = 2\right)$$

$$= P_\theta\left(\frac{X_1}{X_{(3)}} \le \frac{x}{2}\right) \quad \text{(Basu's Thm)}$$

$$= 1 - P_\theta\left(X_{(3)} < \frac{2X_1}{x}\right)$$

$$= 1 - P_\theta\left(X_2 < \frac{2X_1}{x},\ X_3 < \frac{2X_1}{x}\right)$$

$$= 1 - \int_0^1 \left(P\left(X_2 < \frac{2t}{x}\right)\right)^2 dt$$

$$= 1 - \left[\int_0^{x/2} \frac{4t^2}{x^2}\,dt + \int_{x/2}^1 dt\right]$$

$$= 1 - \left[\frac{4}{x^2}\cdot\frac{1}{3}\cdot\frac{x^3}{8} + 1 - \frac{x}{2}\right]$$

$$= \frac{x}{2} - \frac{x}{6} = \frac{x}{3}.$$

$$P_\theta(X_1 \le x \mid X_{(3)} = 2) = \begin{cases} 0, & x < 0 \\ \frac{x}{3} & 0 \le x < 2 \\ 1 & x \ge 2 \end{cases}$$

$\hookrightarrow$ discontinuous at $x = 2$ and continuous everywhere with $P_\theta(X_1 = 2 \mid X_{(3)} = 2) = \frac{1}{3}$

$\longrightarrow$ Mixed type dist.

... 6 MARKS

The derived conditional d.f. is

$$G(x) = \frac{1}{3} F_d(x) + \frac{2}{3} F_c(x), \quad x \in \mathbb{R},$$

where

$$F_d(x) = \begin{cases} 0 & x < 2 \\ 1 & x \geq 2 \end{cases}; \quad F_c(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 2 \\ 1 & x \geq 2 \end{cases}.$$

(b) $X_{(3)}$ is complete-sufficient and $\frac{X_{(1)}}{X_{(3)}}$ is ancillary, so they are independent by Basu's Theorem. For evaluation of $E_\theta\left(\frac{X_{(1)}}{X_{(3)}}\right)$ WLOG we may assume $\theta = 1$, as $\frac{X_{(1)}}{X_{(3)}}$ is ancillary. Then

$$E_\theta(X_{(1)}) = E_\theta\left(X_{(3)} \cdot \frac{X_{(1)}}{X_{(3)}}\right)$$

$$= E_\theta(X_{(3)})\, E_\theta\left(\frac{X_{(1)}}{X_{(3)}}\right) \quad \text{(Basu's Thm)}$$

... $\boxed{\text{3 MARKS}}$

$$\Rightarrow E_\theta\left(\frac{X_{(1)}}{X_{(3)}}\right) = \frac{E_\theta(X_{(1)})}{E_\theta(X_{(3)})} = \frac{\int_0^1 x \, 3(1-x)^2 \, dx}{\int_0^1 x \cdot 3x^2 \, dx}$$

$$= \frac{1}{3}.$$

... $\boxed{\text{3 MARKS}}$

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